

AWS Task-4 – Sudir

Task Description:

Launch an EC2 instance (Linux and Windows) along with a web server. Then, create an EBS volume of 5 GB, attach it to an EC2 machine (Linux and Windows), and take a snapshot. Finally, create an EBS volume using the taken snapshot.

Techstacks needs to be used :

- AWS EBS
- AWS EC2

How do I submit my work?

- Push all your work files to GitHub (O/P screenshot images must).
- Submit your URLs in the portal.

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- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

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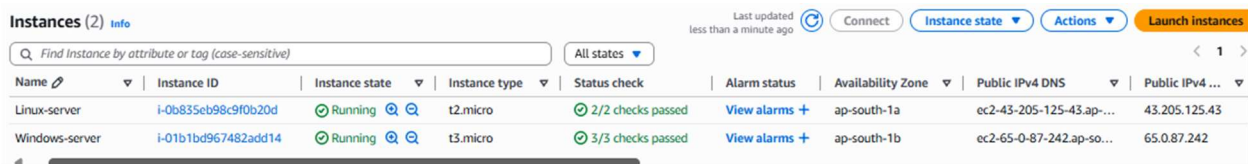
1. Launching EC2 Instances

Explanation:

Two EC2 instances were launched:

- A **Linux EC2 (Amazon Linux 2)** for hosting Apache web server.
 - A **Windows EC2 (Windows Server 2022)** for hosting IIS web server.
- Security groups were configured to allow HTTP, SSH (Linux), and RDP (Windows) access.

Screenshot:



The screenshot shows the AWS Management Console 'Instances' page. It displays two EC2 instances: 'Linux-server' and 'Windows-server'. Both are in a 'Running' state. The 'Linux-server' is an 't2.micro' instance with 2/2 checks passed. The 'Windows-server' is a 't3.micro' instance with 3/3 checks passed. Both instances are in the 'ap-south-1a' availability zone. The 'Linux-server' has a public IPv4 address of 43.205.125.43. The 'Windows-server' has a public IPv4 address of 65.0.87.242.

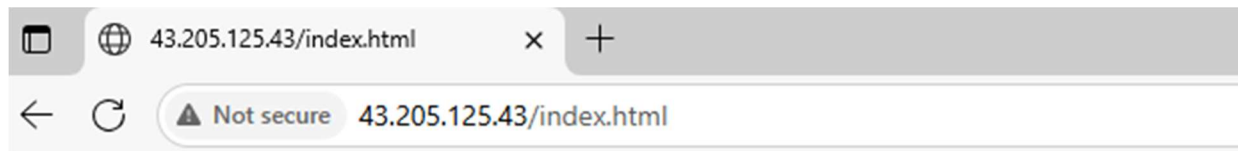
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
Linux-server	i-0b835eb98c9f0b20d	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-43-205-125-43.ap-...	43.205.125.43
Windows-server	i-01b1bd967482add14	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-65-0-87-242.ap-so...	65.0.87.242

2. Configuring Web Server on Linux

Explanation:

On the Linux EC2 instance, Apache (httpd) was installed and configured. A test HTML page was created to verify the setup.

Screenshot:



Hello from Linux EC2 Web Server

3. Configuring Web Server on Windows

Explanation:

On the Windows EC2 instance, IIS was installed using Server Manager. A test HTML page was placed inside C:\inetpub\wwwroot\ to confirm the setup.

Screenshot:



Root volumes of Linux & Windows server

Volumes (2) [Info](#)

Last updated less than a minute ago [Actions](#) [Create volume](#)

Save filter sets [Choose filter set](#)

☐	Name ↗	Volume ID ↗	Type ↗	Size ↗	IOPS ↗	Throughput ↗	Snapshot ID ↗	Created ↗	Availability Zone ↗
☐	linux-root-volume	vol-0aab622501139cee5	gp3	8 GiB	3000	125	snap-0aef119f...	2025/08/25 17:16 GMT+5:...	ap-south-1a
☐	windows-root-volume	vol-055b2f25e8582decf	gp3	30 GiB	3000	125	snap-0e7b637...	2025/08/25 17:18 GMT+5:...	ap-south-1b

4. Creating and Attaching EBS Volumes

Explanation:

A **5 GB EBS volume** was created for each instance (Linux and Windows). Volumes were attached to their respective instances in the same Availability Zone.

Linux

ADVANCED FILTERS

Choose filter set

Search

< 1 >

☒	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone
☒	linux-data-volume	vol-06cc938c472a00466	gp3	5 GiB	3000	125	-	2025/08/25 17:26 GMT+5:...	ap-south-1a
☐	linux-root-volume	vol-0aab622501139cee5	gp3	8 GiB	3000	125	snap-0aef119f...	2025/08/25 17:16 GMT+5:...	ap-south-1a

Windows

☐	windows-root-volume	vol-055b2f25e8582decf	gp3	30 GiB	3000	125	snap-0e7b637...	2025/08/25 17:18 GMT+5:...	ap-south-1b
☐	windows-data-volume	vol-04c09c1cd11cdd37f	gp3	5 GiB	3000	125	-	2025/08/25 17:26 GMT+5:...	ap-south-1b

Data volume attached for Linux server

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch i

Find Instance by attribute or tag (case-sensitive)

All states

Instance ID = i-0b835eb98c9f0b20d

Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Pu
☒	Linux-server	i-0b835eb98c9f0b20d	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-43-205-125-43.ap-...	43

i-0b835eb98c9f0b20d (Linux-server)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Root device details

Root device name

/dev/xvda

Root device type

EBS

EBS optimization

disabled

▼ Block devices

Filter block devices

☒	Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	Attachment time	Encrypted	KMS key ID
☒	vol-0aab622501139cee5	/dev/xvda	8	In-use	Attached	2025/08/25 17:16 GMT+5:30	No	-
☐	vol-06cc938c472a00466	/dev/sdb	5	In-use	Attached	2025/08/25 17:27 GMT+5:30	No	-

Data volume attached for Windows server

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Instance ID = i-01b1bd967482add14 Clear filters

Windows-server i-01b1bd967482add14 Running t3.micro 3/3 checks passed View alarms + ap-south-1b ec2-65-0-87-242.ap-so... 65.0.87.242

i-01b1bd967482add14 (Windows-server)

Details Status and alarms Monitoring Security Networking Storage Tags

Root device details

Root device name /dev/sda1 Root device type EBS EBS optimization enabled

Block devices

Volume ID	Device name	Volume size (GiB)	Volume State	Attachment status	Attachment time	Encrypted	KMS key ID
vol-055b2725e8582decf	/dev/sda1	30	In-use	Attached	2025/08/25 17:18 GMT+5:30	No	-
vol-04c09c1cd11cdd37f	xvdb	5	In-use	Attached	2025/08/25 17:27 GMT+5:30	No	-

5. Mounting and Formatting EBS Volume on Linux

Explanation:

The attached Linux volume /dev/xvdb was formatted with **XFS filesystem** and mounted on /data_volume.

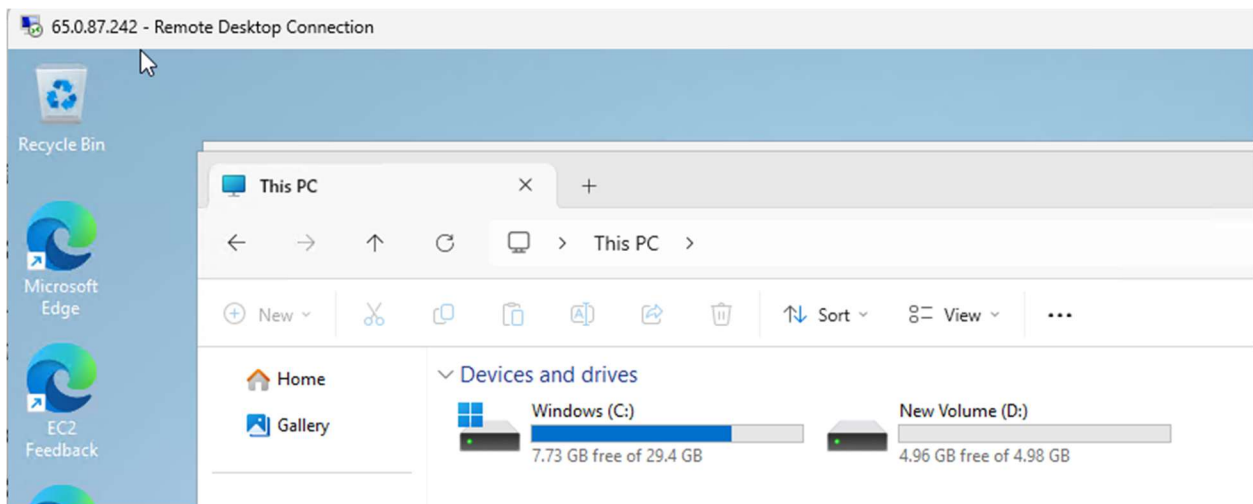
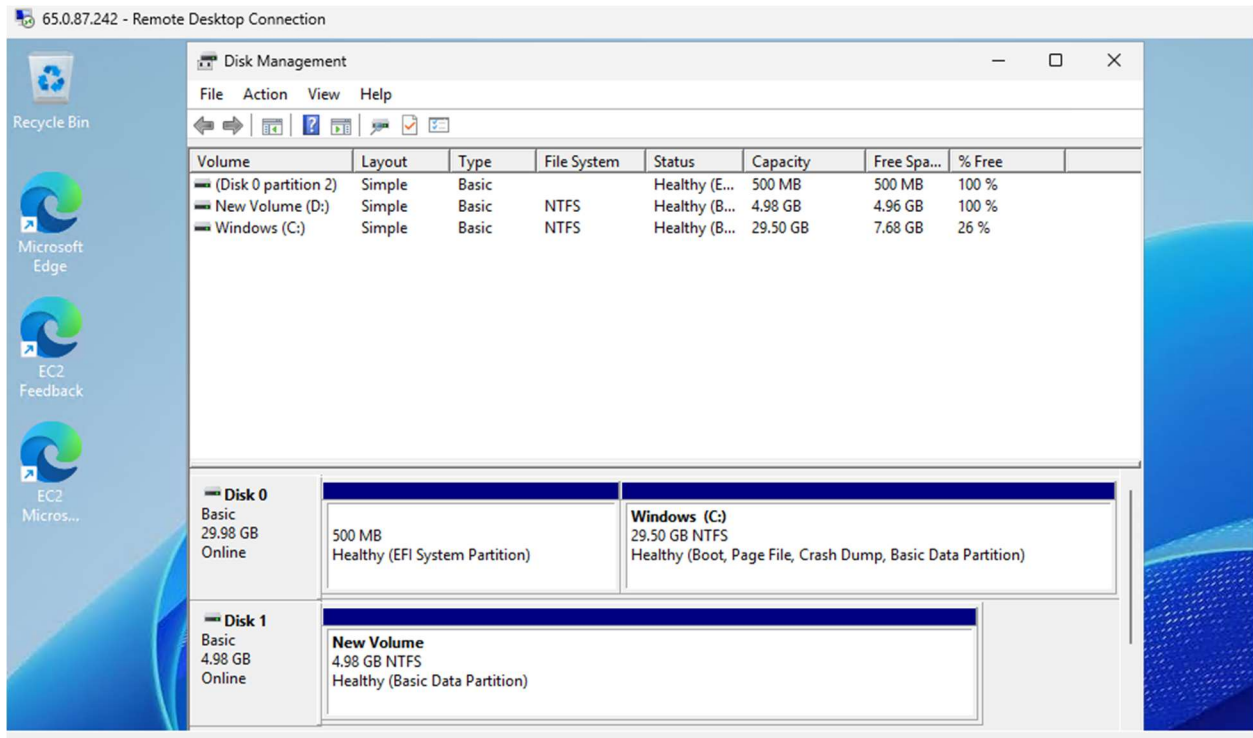
Screenshot:

```
[root@ip-172-31-40-209 ~]# df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
devtmpfs        devtmpfs  4.0M   0  4.0M   0% /dev
tmpfs           tmpfs     475M   0  475M   0% /dev/shm
tmpfs           tmpfs     190M 452K  190M   1% /run
/dev/xvda1      xfs       8.0G  1.6G   6.4G  20% /
tmpfs           tmpfs     475M   0  475M   0% /tmp
/dev/xvda128    vfat      10M   1.3M   8.7M  13% /boot/efi
tmpfs           tmpfs     95M    0   95M   0% /run/user/1000
[root@ip-172-31-40-209 ~]# sudo mkfs -t xfs /dev/xvdb
meta-data=/dev/xvdb          isize=512    agcount=4, agsize=327680 blks
=                               sectsz=512   attr=2, projid32bit=1
=                               crc=1      finobt=1, sparse=1, rmapbt=0
=                               reflink=1  bigtime=1 inobtcount=1
data      =                bsize=4096   blocks=1310720, imaxpct=25
=                               sunit=0    swidth=0 blks
naming    =version 2      bsize=4096   ascii-ci=0, ftype=1
log       =internal log bsize=4096   blocks=16384, version=2
=                               sectsz=512   sunit=0 blks, lazy-count=1
realtime  =none        extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-40-209 ~]# sudo mkdir /data_volume
[root@ip-172-31-40-209 ~]# sudo mount /dev/xvdb /data_volume
[root@ip-172-31-40-209 ~]# echo "/dev/xvdb /data_volume xfs defaults,nofail 0 2" | sudo tee -a /etc/fstab
/dev/xvdb /data_volume xfs defaults,nofail 0 2
[root@ip-172-31-40-209 ~]# cat /etc/fstab
#
UUID=62c90582-6d8a-4bf0-92e8-95e89689ef33 / xfs defaults,noatime 1 1
UUID=E565-8542 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2
/dev/xvdb /data_volume xfs defaults,nofail 0 2
[root@ip-172-31-40-209 ~]# df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
devtmpfs        devtmpfs  4.0M   0  4.0M   0% /dev
tmpfs           tmpfs     475M   0  475M   0% /dev/shm
tmpfs           tmpfs     190M 448K  190M   1% /run
/dev/xvda1      xfs       8.0G  1.6G   6.4G  20% /
tmpfs           tmpfs     475M   0  475M   0% /tmp
/dev/xvda128    vfat      10M   1.3M   8.7M  13% /boot/efi
tmpfs           tmpfs     95M    0   95M   0% /run/user/1000
/dev/xvdb       xfs       5.0G   68M   4.9G   2% /data_volume
[root@ip-172-31-40-209 ~]#
```

Mounting and Formatting EBS Volume on Windows

Explanation:

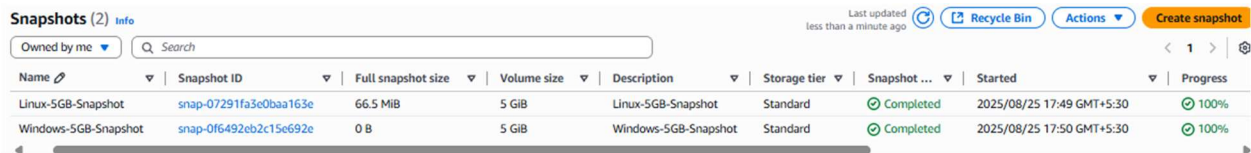
On the Windows EC2 instance, diskmgmt.msc was used to initialize and format the attached disk. The disk was assigned drive letter **D:** and formatted with NTFS.



7. Creating Snapshots

Explanation:

Snapshots were created for both Linux and Windows EBS volumes from the AWS Management Console. These snapshots serve as backups of the volumes.



The screenshot shows the AWS Snapshots page with two snapshots listed. The first snapshot is 'Linux-5GB-Snapshot' with ID 'snap-07291fa3e0baa163e', a size of 66.5 MiB, and a volume size of 5 GiB. It is in a 'Completed' state. The second snapshot is 'Windows-5GB-Snapshot' with ID 'snap-0f6492eb2c15e692e', a size of 0 B, and a volume size of 5 GiB. It is also in a 'Completed' state. Both snapshots were created on 2025/08/25 at 17:49 GMT+5:30 and 17:50 GMT+5:30 respectively, with 100% progress.

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot ...	Started	Progress
Linux-5GB-Snapshot	snap-07291fa3e0baa163e	66.5 MiB	5 GiB	Linux-5GB-Snapshot	Standard	Completed	2025/08/25 17:49 GMT+5:30	100%
Windows-5GB-Snapshot	snap-0f6492eb2c15e692e	0 B	5 GiB	Windows-5GB-Snapshot	Standard	Completed	2025/08/25 17:50 GMT+5:30	100%

7. Creating volumes from the snapshot

I know how to create volumes from these snapshots

Select the snapshot -> Actions -> Create volume.

I have also created the volumes from the above-mentioned snapshots.

But before taking screenshot the Aws lab suddenly disconnected and it shows Deleted.

